

## CAPE HENRY LH, VA, USA

WMO: 997691

Lat: 36.9271N

Lon: 76.0066W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|--------------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                          |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>22.1 | (c)<br>26.1 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>38.4           | (k)<br>29.3 | (l)<br>35.0 | (m)<br>30.4 | (n)<br>18.9              | (o)<br>330 | (p)<br>0.543 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>9.4             | (c)<br>90.7     | (d)<br>N/A | (e)<br>87.9 | (f)<br>N/A | (g)<br>85.6 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>10.1          | (p)<br>220 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 35.1          | (b) 30.7 | (c) 26.7 | (e) 19.5                   | (f) 95.5 | (g) 5.2            | (h) 1.2 | (i) 15.7                                           | (j) 96.3 | (k) 12.7   | (l) 97.0 | (m) 9.8    | (n) 97.7 | (o) 6.0    | (p) 98.5 |  |  |
|                   |          |          | DB                         | N/A      | N/A                | N/A     | N/A                                                | N/A      | N/A        | N/A      | N/A        | N/A      | N/A        | N/A      |  |  |
|                   |          |          | WB                         | N/A      | N/A                | N/A     | N/A                                                | N/A      | N/A        | N/A      | N/A        | N/A      | N/A        | N/A      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 60.9   | 41.5  | 42.8  | 48.5  | 59.1  | 67.1  | 75.4  | 78.6  | 77.4  | 73.0  | 64.5  | 53.3  | 48.5  |  |  |
|                                                                        | DBStd       | 15.00  | 9.93  | 9.37  | 9.04  | 8.14  | 6.88  | 5.26  | 3.98  | 3.26  | 4.42  | 6.77  | 7.07  | 9.05  |  |  |
|                                                                        | HDD50       | 885    | 302   | 242   | 144   | 13    | 0     | 0     | 0     | 0     | 0     | 1     | 46    | 137   |  |  |
|                                                                        | HDD65       | 3111   | 730   | 625   | 517   | 216   | 59    | 2     | 0     | 0     | 1     | 90    | 355   | 516   |  |  |
|                                                                        | CDD50       | 4867   | 38    | 39    | 96    | 285   | 531   | 763   | 887   | 850   | 691   | 452   | 144   | 91    |  |  |
|                                                                        | CDD65       | 1615   | 0     | 2     | 4     | 38    | 125   | 315   | 422   | 385   | 242   | 75    | 3     | 4     |  |  |
|                                                                        | CDH74       | 10657  | 0     | 5     | 22    | 235   | 620   | 2235  | 3516  | 2751  | 1052  | 208   | 5     | 8     |  |  |
|                                                                        | CDH80       | 2591   | 0     | 0     | 0     | 48    | 134   | 648   | 1009  | 563   | 164   | 25    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 12.3   | 13.3  | 13.1  | 14.1  | 12.8  | 11.6  | 10.7  | 9.5   | 9.7   | 12.3  | 13.4  | 14.3  | 12.6  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 46.5   | 3.0   | 2.9   | 3.4   | 3.2   | 3.3   | 4.3   | 5.6   | 5.4   | 5.2   | 3.9   | 3.1   | 3.3   |  |  |
|                                                                        | PrecMax     | 59.6   | 6.1   | 8.2   | 9.0   | 6.1   | 5.5   | 11.3  | 9.0   | 13.0  | 15.4  | 8.4   | 11.2  | 7.0   |  |  |
|                                                                        | PrecMin     | 28.1   | 0.9   | 0.6   | 0.5   | 0.5   | 0.9   | 1.6   | 0.7   | 1.0   | 0.3   | 0.2   | 0.1   | 1.5   |  |  |
|                                                                        | PrecStd     | 7.9    | 1.3   | 1.6   | 1.6   | 1.3   | 1.1   | 2.0   | 2.2   | 3.0   | 3.7   | 2.1   | 2.3   | 1.5   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 70.8  | 74.5  | 77.5  | 85.4  | 87.5  | 93.6  | 93.8  | 92.2  | 89.0  | 83.6  | 74.7  | 75.1  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | DB     | 67.1  | 68.9  | 71.6  | 80.3  | 83.9  | 89.9  | 90.8  | 88.5  | 84.5  | 78.9  | 69.8  | 71.0  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | DB     | 62.1  | 64.8  | 67.8  | 76.1  | 80.5  | 86.5  | 87.9  | 85.4  | 81.3  | 76.1  | 66.5  | 66.6  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | DB     | 57.7  | 58.3  | 63.5  | 71.7  | 77.3  | 83.6  | 85.4  | 82.9  | 79.1  | 73.8  | 63.6  | 62.8  |  |  |
|                                                                        |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.7  | 12.4  | 12.0  | 12.8  | 10.8  | 9.9   | 9.4   | 7.8   | 7.1   | 9.1   | 9.5   | 10.6  |  |  |
|                                                                        |             | MCDBR  | 19.6  | 20.4  | 20.8  | 20.4  | 16.8  | 15.2  | 14.5  | 13.2  | 12.2  | 12.8  | 14.7  | 16.9  |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
|                                                                        |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.312 | 0.322 | 0.348 | 0.393 | 0.437 | 0.484 | 0.512 | 0.505 | 0.410 | 0.357 | 0.326 | 0.316 |  |  |
|                                                                        | taud        |        | 2.545 | 2.509 | 2.468 | 2.354 | 2.263 | 2.183 | 2.117 | 2.123 | 2.382 | 2.522 | 2.555 | 2.562 |  |  |
|                                                                        | Ebn at Noon |        | 281   | 291   | 291   | 282   | 270   | 256   | 248   | 246   | 267   | 274   | 272   | 272   |  |  |
|                                                                        | Edh at Noon |        | 24    | 29    | 33    | 39    | 43    | 47    | 49    | 48    | 35    | 28    | 24    | 23    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 733   | 980   | 1318  | 1682  | 1896  | 2035  | 1942  | 1752  | 1432  | 1129  | 831   | 655   |  |  |
|                                                                        | RadStd      |        | 59    | 101   | 102   | 118   | 160   | 130   | 93    | 104   | 112   | 121   | 66    | 51    |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (52 neighbors) |  | +0.79 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +295  | +187  |

Nomenclature: See separate page